TEXORA · LECTURE SHEET

Fluid Mechanics

The physics of liquids and gases in motion: properties and pressure, continuity of flow, Bernoulli's energy equation and its meters, laminar versus turbulent flow, and the friction that every pipeline quietly charges — the groundwork for pumps, dye liquor circulation and air handling.

ME-3203 · THERMAL ENGINEERING AND FLUID MECHANICS · TOPIC 3 · LEVEL 3 TERM 2

1. Properties and pressure

A fluid deforms continuously under any shear stress — that single fact separates fluids from solids. The working properties: **density** ρ (water ≈ 1000, air ≈ 1.2 kg/m³), **viscosity** μ, the internal friction that resists shearing (Newton's law $\tau = \mu\, du/dy$), and its kinematic form $\nu = \mu/\rho$. Viscosity is why cold dye liquor circulates reluctantly and why heating a bath makes pumping cheaper: water at 80 °C is barely one-third as viscous as at 20 °C.

Pressure in a static liquid grows linearly with depth, $p = \rho g h$ (the **hydrostatic law**), and **Pascal's law** transmits pressure undiminished in all directions — the principle of every hydraulic press and of the even squeeze in a pad mangle. Pressure is quoted as **gauge** (above atmosphere) or **absolute**; pump suction calculations (Topic 4) live or die on keeping the two straight. Manometers — U-tube and inclined — convert pressure differences into readable liquid columns.

2. Continuity — what flows in, flows out

For steady flow of an incompressible fluid the volume rate Q is the same at every section:

$$Q = A_1 v_1 = A_2 v_2$$

Squeeze the pipe and the fluid speeds up in exact inverse proportion to area. The worked pipe of this sheet: a 150 mm main at 2 m/s carries $Q = (\pi/4)(0.15)^2 \times 2 =$ **0.0353 m³/s ≈ 35 L/s**. Branching obeys the same ledger (inflow = sum of outflows), which is how liquor distribution headers and air ducts are balanced.

3. Bernoulli — the energy ledger of a flowing stream

Along a streamline, for steady incompressible frictionless flow, the total head is constant:

$$p/\rho g + v^2/2g + z = \text{constant (total head, metres)}$$

Pressure head, velocity head and elevation head trade against each other: speed the fluid up and its pressure falls. That trade runs the **venturi meter** (a smooth constriction whose pressure drop measures flow), the **pitot tube** (stagnation pressure measures velocity, standard for duct air surveys) and the carburettor — and it explains suction at contractions everywhere in plant pipework.

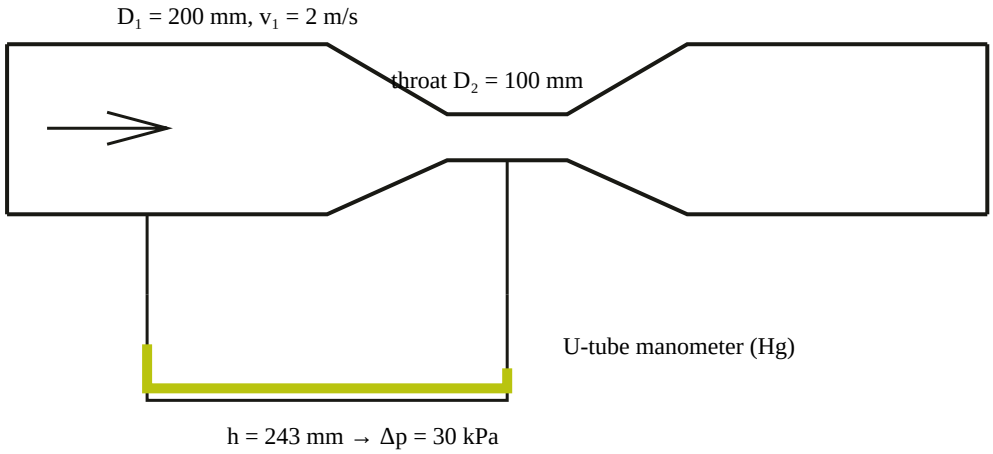


Figure 1 — Venturi meter with differential U-tube manometer. Continuity quadruples the throat velocity; Bernoulli converts the speed into a measurable pressure drop.

WORKED EXAMPLE — VENTURI METER

Water flows through a 200 mm pipe fitted with a 100 mm throat venturi; upstream velocity is 2 m/s. Continuity: $v_2 = v_1(A_1/A_2) = 2 \times (200/100)^2 =$ **8 m/s**. Bernoulli: $p_1 - p_2 = \tfrac{1}{2}\rho(v_2^2 - v_1^2) = \tfrac{1}{2} \times 1000 \times (64 - 4) =$ **30 kPa**. On a mercury U-tube across the tappings this shows $h = \Delta p/((\rho_Hg - \rho)g) = 30{,}000/(12{,}600 \times 9.81) =$ **0.243 m ≈ 243 mm** — a comfortable reading, which is exactly why mercury (13.6× water's density) is chosen: water itself would need a 3 m column.

4. Laminar, turbulent, and the Reynolds number

Flow organises itself by the **Reynolds number**, the ratio of inertia to viscous forces:

$$\mathrm{Re} = vD/\nu$$

Below ≈ 2000 flow in a pipe is **laminar** — smooth layers, parabolic velocity profile, friction proportional to velocity. Above ≈ 4000 it is **turbulent** — eddying, well-mixed, flatter profile, friction roughly proportional to velocity squared. The pipe flow of Section 2 at 20 °C ($\nu = 1.0 \times 10^{-6}$ m²/s): $\mathrm{Re} = 2 \times 0.15/1.0 \times 10^{-6} \approx$ **3.0×10^5** — deep in turbulence, as nearly all mill water flow is. Laminar flow

survives in viscous oils, polymer melts and slow seepage through fabric — where its orderly layers are exactly what dye penetration and filtration theory need.

5. Friction in pipes — Darcy–Weisbach

Real flow pays a toll. The head lost to wall friction over a pipe run is:

$$h_f = f\,(L/D)\,(v^2/2g)$$

with friction factor f from the Moody chart as a function of Re and wall roughness (smooth drawn pipe ≈ 0.015–0.02 at high Re; corroded steel much worse). **Minor losses** — bends, valves, entries, exits — are charged as $K(v^2/2g)$ each, or folded into an equivalent length.

> **WORKED EXAMPLE — PIPELINE FRICTION**
>
> The 150 mm main carries water at 2 m/s over 100 m with f = 0.02: $h_f = 0.02 \times (100/0.15) \times (2^2/(2 \times 9.81)) = 0.02 \times 667 \times 0.204 =$ **2.72 m of head** — about 27 kPa surrendered to friction, before a single bend or valve is counted. Double the velocity to force more production and the toll quadruples (v^2 law) to nearly 11 m — the quiet arithmetic behind every “just run the pump faster” argument, and behind Topic 4’s system curve.

6. Key points

- Continuity: Q = Av everywhere; constriction multiplies velocity by the area ratio (2 → 8 m/s in the venturi).
- Bernoulli: pressure, velocity and elevation heads trade to a constant total; the venturi drop of 30 kPa reads 243 mm on mercury.
- Re = vD/ν sorts laminar (< 2000) from turbulent (> 4000); mill water is turbulent, oils and melts are laminar.
- Darcy–Weisbach: $h_f = f(L/D)(v^2/2g)$ — 2.72 m per 100 m in the example, and it grows with v^2.
- Gauge vs absolute pressure discipline is essential before any suction-side calculation.

7. Practice

Answer in your own words; brief model answers follow.

1. A 100 mm pipe branches into two 50 mm pipes. If the main carries 1.6 m/s, and the branches share the flow equally, find each branch velocity.
2. A pitot tube in a water stream reads a stagnation pressure 4.9 kPa above the static pressure. Find the stream velocity.

3. Oil ($\nu = 1.0 \times 10^{-4}$ m²/s) flows at 0.5 m/s in a 50 mm line. Find Re and the flow regime, and state the profile shape.
4. For the friction example, add six bends with K = 0.9 each and a fully open gate valve K = 0.15. Find the extra head loss.
5. A tank's free surface is 4 m above a tap. Using Bernoulli (Torricelli), find the ideal exit velocity and the discharge through a 20 mm tap if a discharge coefficient of 0.8 applies.
6. Why does heating dye liquor from 20 °C to 80 °C reduce pumping power even at the same volume flow?

Answer key

1. $Q = (\pi/4)(0.1)^2 \times 1.6 = 0.01257$ m³/s; each branch takes 0.00628 m³/s through $(\pi/4)(0.05)^2 = 0.00196$ m² → v = 3.2 m/s. (Equal areas in total to half the main's area, hence double velocity, split equally.)
2. $v = \sqrt{(2\Delta p/\rho)} = \sqrt{(2 \times 4900/1000)} = \sqrt{9.8} = 3.13$ m/s.
3. $Re = 0.5 \times 0.05/1.0 \times 10^{-4} = 250$ — laminar. Profile is parabolic, centre velocity twice the mean.
4. Velocity head $= v^2/2g = 0.204$ m. $\Sigma K = 6 \times 0.9 + 0.15 = 5.55$. Extra loss $= 5.55 \times 0.204 = 1.13$ m — fittings add ~40% on top of straight-pipe friction here.
5. Ideal $v = \sqrt{(2gh)} = \sqrt{(2 \times 9.81 \times 4)} = 8.86$ m/s. $Q = 0.8 \times (\pi/4)(0.02)^2 \times 8.86 = 0.8 \times 3.14 \times 10^{-4} \times 8.86 = 2.23 \times 10^{-3}$ m³/s ≈ 2.2 L/s.
6. Viscosity falls roughly threefold ($1.0 \rightarrow 0.36 \times 10^{-6}$ m²/s kinematic), so Re rises, friction factor falls, and in any laminar sections friction collapses outright; the system curve flattens, the duty point shifts to higher flow at lower head, and shaft power ($\rho g Q H/\eta$) drops accordingly.
